



Open Automated Driving

Stack | Sim | Suite | Safety

IEEE ITSC 2026 | TUTORIAL



thinking cars

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RWTH AACHEN
UNIVERSITY



11:30 – 11:35	Welcome and Introduction Scope, learning objectives, and setup. · All organizers
11:35 – 11:45	Why Collaborative OpenADS Development? Timo Woopen, Thinking Cars GmbH
11:45 – 12:20	OpenADS Reference Architecture Raphael van Kempen, RWTH Aachen University
12:20 – 13:00	OpenADS in Practice: Ecosystem Applications Data Foundations Benchmark-driven Qualification of OSS-Modules OSS-based Fleet Management
13:00 – 14:00	Lunch Break
14:00 – 14:50	OpenADS Hands-On Christian Geller and Raphael van Kempen, RWTH Aachen University
14:50 – 15:00	Wrap-Up and Summary Lessons learned and next collaboration steps. All organizers



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Why Collaborative OpenADS Development?

Dr.-Ing. Timo Wooten

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Status Quo

Automated Mobility is Reality – in the USA und China

USA

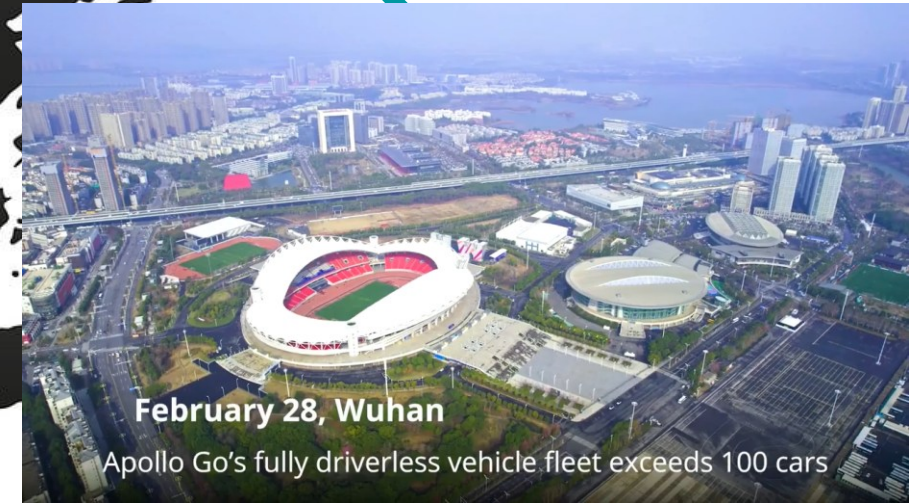
Driverless Robotaxi-Services in commercial operation



Source: <https://www.youtube.com/watch?v=8ebjO1HWQ9Q>

China

Beijings demonstration area alone is 600 km², ~900 vehicles and over 32 Mio. test-km.



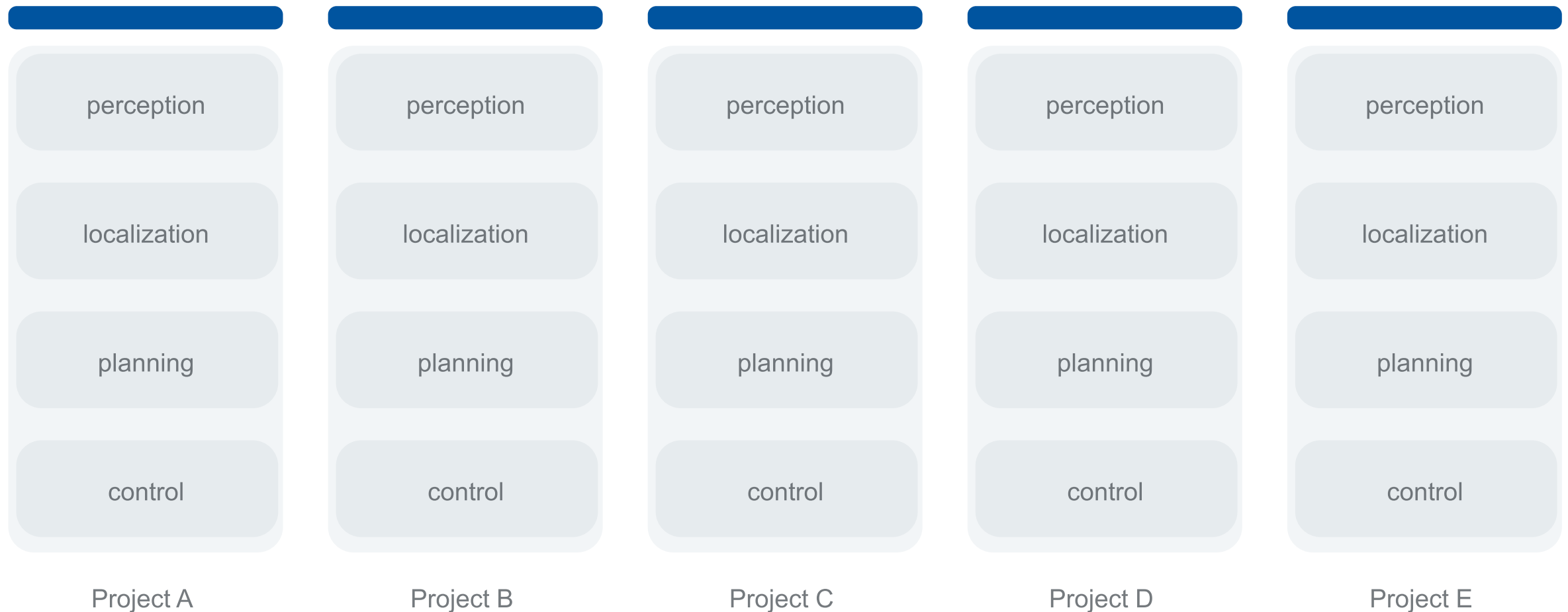
Source: <https://www.youtube.com/watch?v=lrZV4XVaSi4>

Europe

No driverless operation with european AD-tech!

Europe's Challenge is not a capability gap; it's a coordination gap.

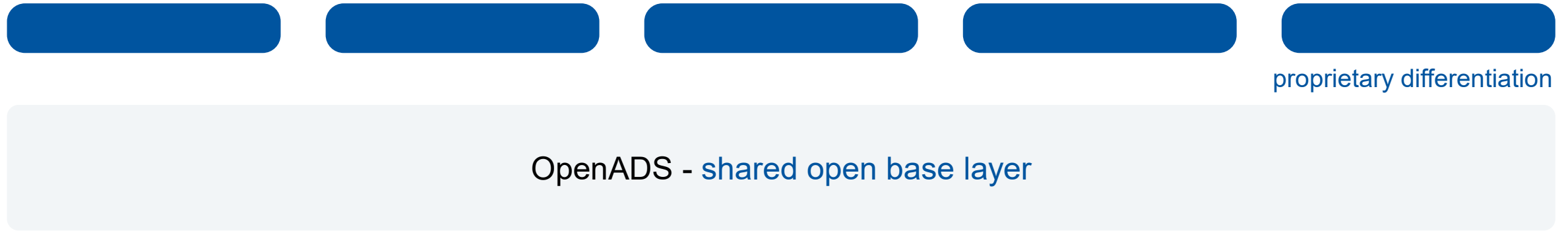
Research output, the suppliers and the safety expertise exist. It lacks is a shared foundation!



Each project rebuilds the same non-differentiating ~80% of the stack.

The thin bar is the differentiating remainder.

Europe's Challenge is not a capability gap; it's a coordination gap.
Share the non-differentiating base, differentiate above it. Under European Sovereignty!



Notable Examples:

CLOUD INFRASTRUCTURE

Linux and Kubernetes became the shared substrate competitors build products on.

ROBOTICS

ROS and ROS 2 gave the field a common interface and message model.

SOFTWARE-DEFINED VEHICLE

Eclipse S-CORE and Eclipse SDV are organising a safety-relevant open base layer in automotive.

European Initiative ECAVA – European Connected and Autonomous Vehicle Alliance



Commission boosts European automotive industry's global competitiveness

Quelle: europa.eu

Source: europa.eu



European Connected and Autonomous Vehicle Alliance (ECAVA)

Terms of Reference

August 2025



ECAVA'S FOCUS IN THE 4 WORKING GROUPS



Software-Defined Vehicles:

Common AI-enabled software tools and open-source building blocks for digital vehicles.



Automotive Hardware:

High-performance computing platforms, AI accelerators, chiplets and open-source standards



AI and Data:

Data pooling, foundational AI models and a European autonomous driving stack.



Autonomous Vehicle Deployment:

Cross-border piloting and real-world deployment on European roads.

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Key deliverables:

The working groups deliver yearly technology, implementation and business roadmaps, endorsed by the Steering Committee, with recommendations that champion industry-driven innovation and foster alignment with the EU and Member States on investment needs and regulation.

German National Initiative Ecosystem Mobility 4.0

1 Community

building sustainable networks and cooperation structures with a genuine sense of belonging

2 Architecture

creating open integration and reference architectures

3 Technology

establishing open technology and methodology building blocks, including open-source solutions

4 Regulation and economy

aligning regulatory and economic frameworks and stimulating a growing provider market

5 Transfer

systematically integrating pilot projects, building knowledge and enabling scalable transfer, both nationally and internationally



Ecosystem Mobility 4.0

115 Follower:innen
1 Monat •

...

Viel Energie, viel Know-how – und ein klares gemeinsames Verständnis:
So lässt sich der gestrige Auftaktworkshop des Ecosystem Mobility 4.0 in Köln gut zusammenfassen. ... mehr



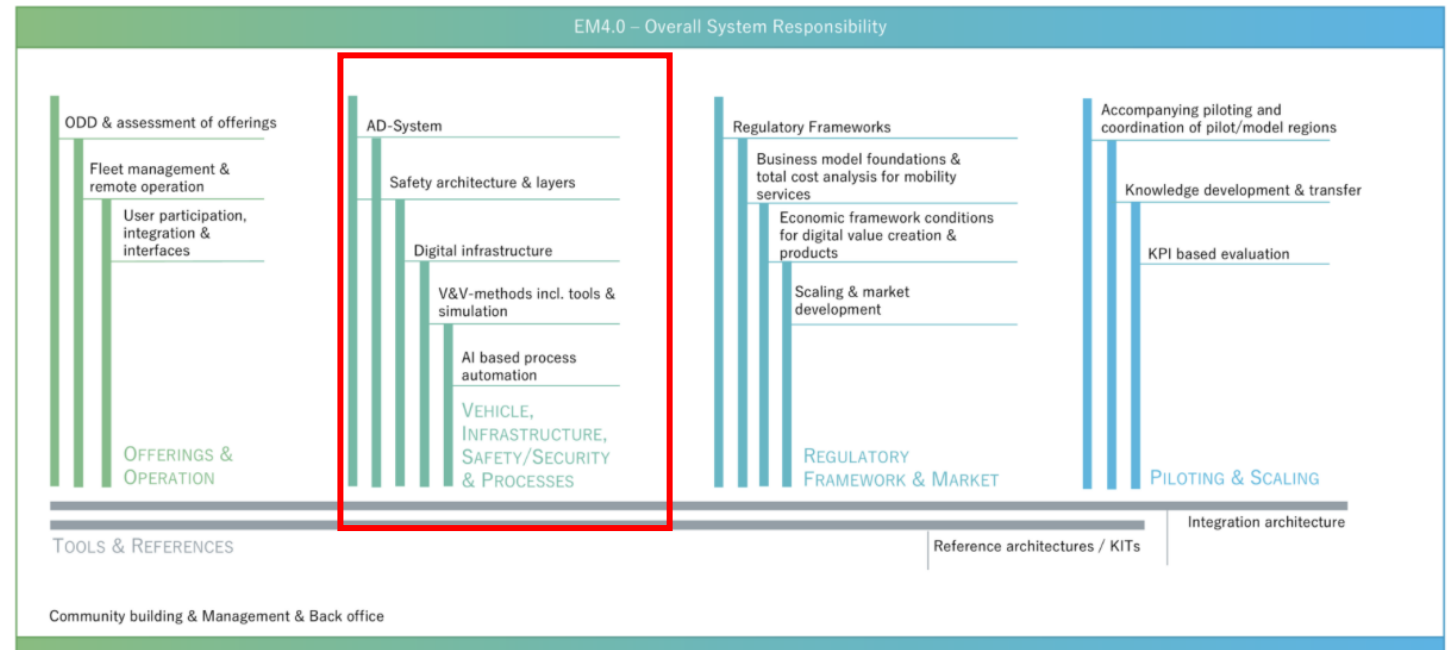
Source: LinkedIn



Source: [Ecosystem Mobility 4.0](#)



Fields of Action



Ecosystem Mobility 4.0 and new industry players supply the missing layers.

01 | GOVERNANCE

Code without governance becomes abandonware.

?

02 | COMPOSABILITY

Code without a shared architecture does not compose.

?

03 | EVIDENCE

Code without reproducible evaluation cannot be integrated by anyone carrying safety responsibility.

?

Ecosystem Mobility 4.0 and new industry players supply the missing layers.

01 | GOVERNANCE

Community building and back-office structures



ECAVA

02 | COMPOSABILITY

Reference architectures and an integration architecture. Vendor neutrality and data sovereignty as design principles



03 | EVIDENCE & INTEGRATION

e.g. Benchmark-driven Qualification of OSS-Modules

Aligned with the European Connected and Autonomous Vehicle Alliance.

Ecosystem Mobility 4.0 and new industry players supply the missing layers.



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02 | COMPOSABILITY

Reference architectures and an integration architecture. Vendor neutrality and data sovereignty as design principles

OpenADS first concrete technical output of the AD-System group

OpenADStack

Modular ROS 2 reference implementation

OpenADServices

Reusable modules: localization, perception, understanding, planning, control

OpenADSim

Closed-loop simulation with CARLA and SUMO

OpenADSuite

Development tooling

OpenADSafety

Benchmarks and V&V tools

Aligned with the European Connected and Autonomous Vehicle Alliance.

Orchestrating Autonomy

Benchmark-driven Qualification of
OSS-Modules





Ulm



Karlsruhe



Aachen



Our founders are part of the UNICARagil and autotech.agil core teams. With their respective institutes they combine extensive knowledge in on- and off-board vehicle intelligence and tooling.

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Why Establish Open-Source Software in Industry?

Ecosystem Advantages on the Big Scale



Speed up development with code-first approach.



Resilient development of OSS excluded from restrictive regulations.

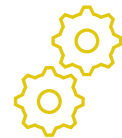


Quality assured by end-to-end control of large community.
e.g. Linux

Individual Economic Advantages

OEM, TIER I and SW-Suppliers can make use of scaling effects by reusing non distinguishing software and focus on relevant tasks.

Service Providers are enabled to offer services to a larger group of customers.



Technological sovereignty is control over the AD roadmap.

If the AD stack sits inside an externally controlled platform, feature quality, feature timing, cost logic, and supplier leverage are often unclear.

Roadmap control
Decide feature priorities and release timing on your own program logic.

Margin leverage
Reduce dependency-driven pricing power and protect long-term economics.

Supplier optionality
Choose modules and partners by fit instead of inherited lock-in.

CURRENT DEPENDENCY PATTERN

External control and dependency today

- Vehicle roadmap
Adaptation speed follows platform boundaries and external priorities.
- Data and diagnostics
Learning loops and debugging depth depend on supplier interfaces.
- Safety evidence
Qualification logic is tied to assumptions you do not fully control.
- Commercial terms
Scope expansion and total cost stay anchored in dependency.



A concrete decision:

Which object detection module do you build on?

No full stack in-house. The vehicle has to be composed from open modules, and each candidate only works in its own demonstration.

Autaware Lidar CenterPoint

Center-based 3D detection, deployed in the Autaware perception pipeline

works in autaware only

FocalFormer3D

Transformer detector with multi-stage focal queries, tuned for hard and missed objects

works in its own demo

PillarOD

Pillar-based one-stage detector on a bird's-eye-view grid

works in its own demo



ONE INTEGRATOR

Picks one, ships it, and carries the consequences for the service life of the vehicle.

Today the honest answer to “which one” is that nobody really knows.

Status Quo of Module Selection:

Modules are selected by reputation, not by evidence.

Autoware Lidar CenterPoint

FocalFormer3D

PillarOD



ONE INTEGRATOR

Demo videos on the maintainer's own recordings

Benchmark numbers reported in papers, on incomparable setups

Personal networks and conference impressions

SAME METRIC NAME, TWO PAPERS

0.12 m

ATE RMSE · MODULE X

Dataset: highway sequences ODD: dry,
daylight Tuning: per-sequence

0.11 m

ATE RMSE · MODULE Y

Dataset: campus loop ODD: mixed
weather Tuning: default parameters



Comparable benchmarks cannot be assembled from published results. They must be designed and executed on one unified platform: [the same scenarios, the same metrics, the same ODD for every module.](#)

Illustrative. The numbers are close; they measure different things.

Benchmark-driven qualification

Reproducible, scenario- & metric-based evidence, generated automatically per module

PROPERTY 1

Comparability

Across contributions and across versions of the same module.

PROPERTY 2

The module is the unit

Qualification applies to the module in a defined deployment, not to the vehicle.

A test tells you it ran. A metric is hard to compare.
Our **benchmark** tells you how it compares, under stated conditions.

Jointly build a sovereign open-source AD stack.

We propose an open marketplace: define, integrate, and qualify the stack you want based on open-source software and generate sovereignty in your software!

DISCOVER

We identify the right open-source components for your target application and compliance requirements and map them against your existing stack.



EVALUATE & QUALIFY

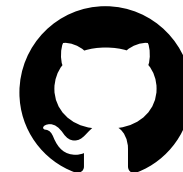
We run reproducible benchmarks across real operating scenarios and safety conditions. Testing performance, safety, and reliability on your data for your auditable evidence.



DEPLOY

We integrate open-source AD-software into your delivery pipelines. We support testing and homologation with our experienced partners.

Ecosystem already growing in OpenADS
<https://github.com/openads-project/>



What changes, by role.

Research

Comparable, citable module results.

OEM

Sovereign make-or-buy decided on measured evidence.

Supplier

An entry ticket without prior OEM qualification.

Integrator

Evidence-based selection with quantified integration risk.

Assessor and
homologation

Traceable evidence feeding ISO 26262 and SOTIF arguments.

What changes, by role.

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Comparable, citable module results.

Three ways to contribute.

A module

Bring an implementation and let CI qualify it.

Support the industrialization and profit in licenses as it gets used in industry!

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A scenario

Add the case your ODD needs and others lack

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A scenario

Add the case your ODD needs and others lack

A metric

Define what your discipline considers evidence

CONTACT ME



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